

Pranav Rathod

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EDUCATION

University of Southern California

Master of Science in Computer Science, Multimedia and Creative Technologies

January 2024 – December 2025

Los Angeles, CA

University of Illinois at Chicago

Bachelor of Science in Computer Science, Minor in Art

August 2019 – May 2023

Chicago, IL

SKILLS

Languages: C/C++, Python, C#, Java, JavaScript, SQL

Graphics & Systems: OpenGL, GLSL, WebGL, Unity, Maya, Multi-threading, Linux/Unix

Tools & Frameworks: Git, Docker, GitHub Actions, PyTorch, Node.js, React, Firebase, GCP

RELEVANT COURSEWORK

Computer Graphics, 3D Graphics and Rendering, Game Engine Development, Computer Animation and Simulation, Robotics, Multimedia Systems Design, Analysis of Algorithms

EXPERIENCE

Software Development Engineer – Mediverse VR Project

February 2026 – Present

Easley Dunn Productions, Inc.

Los Angeles, CA

- Diagnosed CPU-bound bottleneck on Meta Quest by profiling 8-core utilization and GPU time, isolating GameObject overhead over the assumed GPU limit
- Stabilized 68–72 FPS across 3 VR scenes by tuning mesh imports, texture compression, and material parameters on Quest 1
- Authored C# Unity tooling for lighting, object snapping, and scene navigation; resolved a shader bug where base color failed to bind to prefabs

Media Support Specialist

January 2024 – December 2025

Annenberg TechOps Information Technology, University of Southern California

Los Angeles, CA

- Debugged HTML, WordPress, and Adobe Suite issues for 500+ journalism students; provided technical guidance on web design, asset optimization, and code troubleshooting
- Managed student project deployments to Annenberg's server, advising on compression techniques under tight database limits

Research Assistant

June 2022 – December 2023

ELiCIT Lab, University of Illinois at Chicago

Chicago, IL

- Collaborated with Adobe Research to prototype a Flutter-based mobile app for multimodal interaction studies
- Designed UI and backend logic for speech- and gesture-driven image editing across 12 task workflows
- Integrated GCP Speech-to-Text and real-time gesture logging, capturing synchronized multimodal data for NUI training

Teaching Assistant

January 2022 – December 2022

Department of Computer Science, University of Illinois at Chicago

Chicago, IL

- Led discussions and labs on recursion, graph traversal, and dynamic programming for 100+ CS students
- Mentored students through debugging C/C++ assignments and Flutter/Dart application development across 3 semesters

PROJECTS

Autonomous Robotic Navigation & Manipulation Pipeline | Python, ROS2, NumPy

December 2025

- Built a 2000-particle Monte Carlo localizer with LIDAR raycasting and motion-scaled odometry noise
- Extended the filter with a 4-component beam sensor model and K-means pose clustering for multimodal posteriors
- Improved RRT goal precision 5x (0.52 to 0.09 units) via goal-biased sampling at $p=0.2$ with under 30% compute overhead

PrimeEngine | C++, OpenGL, Dear ImGui, Maya

April 2025

- Doubled render throughput from 30 to 60 fps on dense mesh spawns via AABB frustum culling in a custom C++ engine
- Accelerated scene authoring by building a Maya-to-engine pipeline using per-mesh Lua configs to drive runtime mesh loading
- Replaced a broken retained-mode UI with Dear ImGui; debugged a pause-system bug by tracing event-queue ordering across frame boundaries and gating update/physics/scene-graph events

Segmented Video Compression | Python, Detectron2, OpenCV, PyTorch

December 2024

- Implemented a block-based encoder using Three-Step Search motion estimation and per-region DCT quantization
- Strengthened segmentation robustness on moving-camera scenes by combining Detectron2 with Farneback optical flow
- Engineered a synchronized A/V player with real-time IDCT decoding, PyAudio sync, and frame-step controls

ACHIEVEMENTS & LEADERSHIP

SIGGRAPH Student Volunteer Team Leader | Denver, Tokyo, Vancouver, Los Angeles

2024–2026

- Coordinated on-site logistics for international student volunteer teams across program tracks, sessions, and venues
- Onboarded first-time volunteers and resolved live operational issues at world's premier computer graphics conference

Undergraduate Research Forum | Chicago, IL

April 2023

- Achieved 3rd place in the Engineering and Physical Sciences category for presenting NUI-based research